Resampling for limited data

ADVANCED PROBABILITY: UNCERTAINTY IN DATA



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What is resampling?

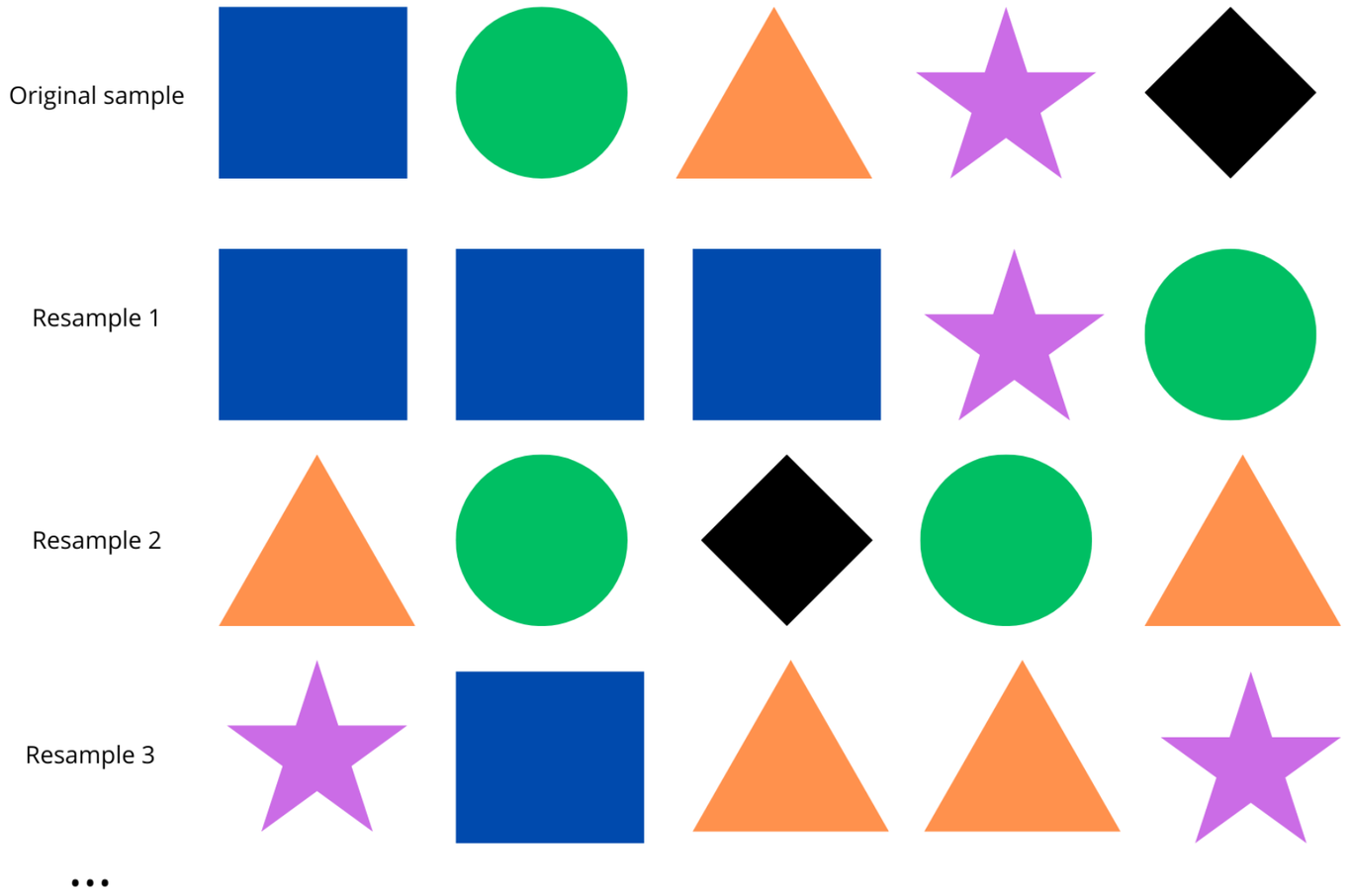
- How many marbles of each color?
- Estimate without counting
- Solution: take repeated samples and record colors



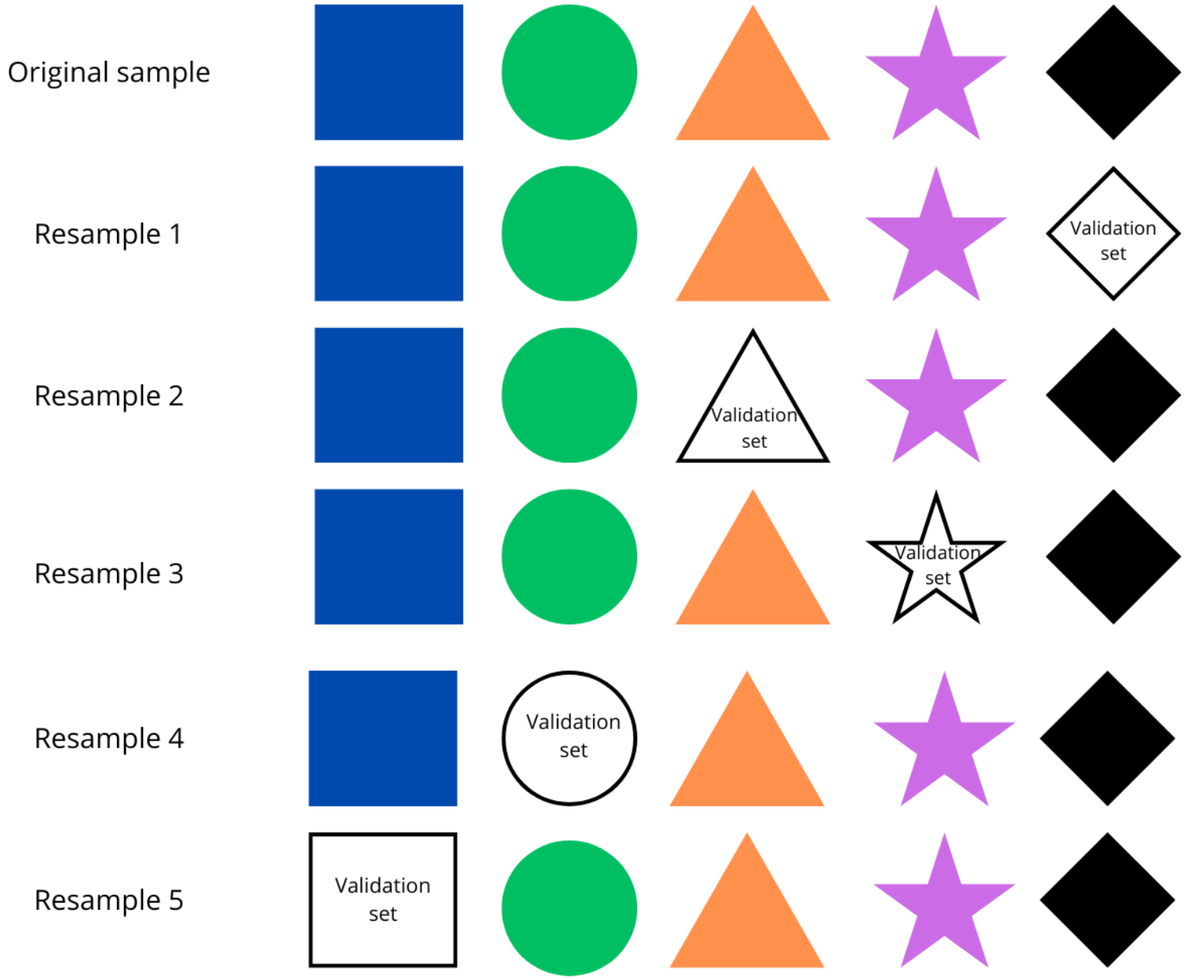
Resampling techniques

- Bootstrapping
- Cross-validation
- Synthetic sampling

Bootstrapping



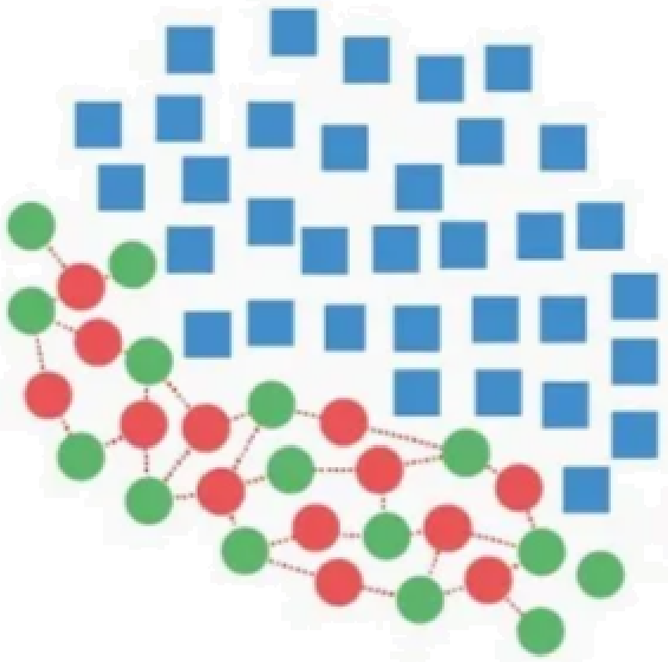
Cross-validation



Synthetic resampling



Original Dataset



Generating Samples



Resampled Dataset

Example: fraud detection

- Fraudulent transactions are rare
  - Models predict all transactions as safe
- Generate synthetic fraud cases
  - Model can learn fraud patterns



Example: fraud detection

- Starting situation:
  - 1,000,000 transactions per month
  - 1,000 or 0.1% classified as fraudulent
- Synthetic sampling:
  - Increase fraudulent samples to 10,000 or 1%



Let's practice!

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Monte Carlo simulation

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What is Monte Carlo simulation?



What is Monte Carlo simulation?

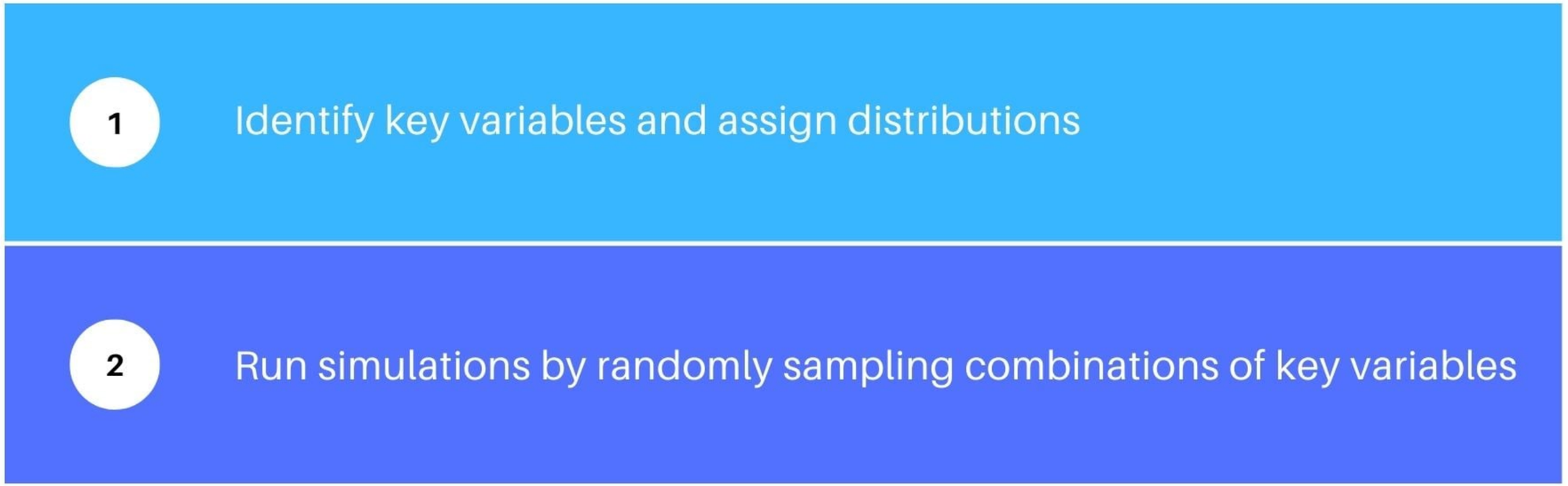
- Try out many random possibilities to see what could happen
- Results in a probability distribution of all simulated outcomes
- Example applications:
  - Financial forecasting
  - Risk assessment
  - Supply chain optimization



Steps in Monte Carlo simulation

1. Identify key variables and assign distributions

Steps in Monte Carlo simulation



Steps in Monte Carlo simulation



Example: project cost estimation

A company wants to estimate total cost of a project

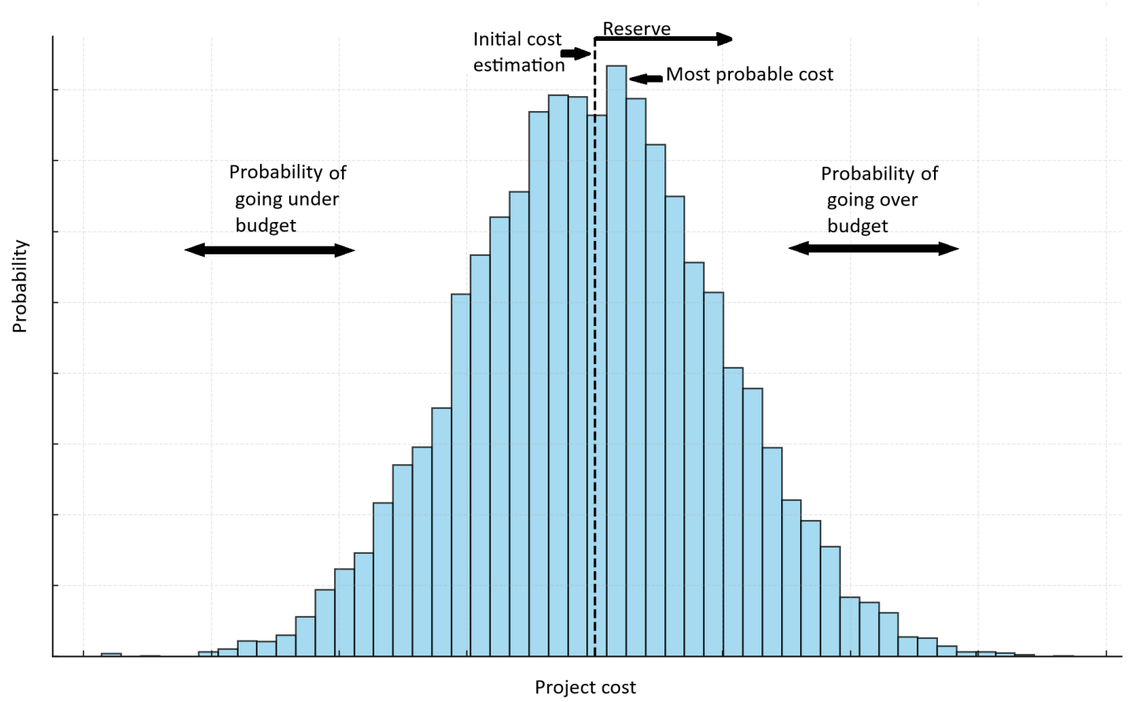
1. Key variables: material prices, labor hours, unexpected expenses



Example: project cost estimation

A company wants to estimate total cost of a project

1. Key variables: material prices, labor hours, unexpected expenses
2. Run 10,000 simulations with randomly sampled values
3. Analyze the results to estimate the probability of staying within budget



Key advantages of Monte Carlo simulation

- Generates a range of possible outcomes
- Quantify risks and opportunities in terms of probabilities
- Capture complexity and interdependencies of variables



Limitations of Monte Carlo simulation

- Accuracy depends on the quality of the input data and assumptions
- Computationally intensive
- Expertise needed to interpret the results



Example: project cost estimation



Let's practice!

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Visualizing paths with decision trees

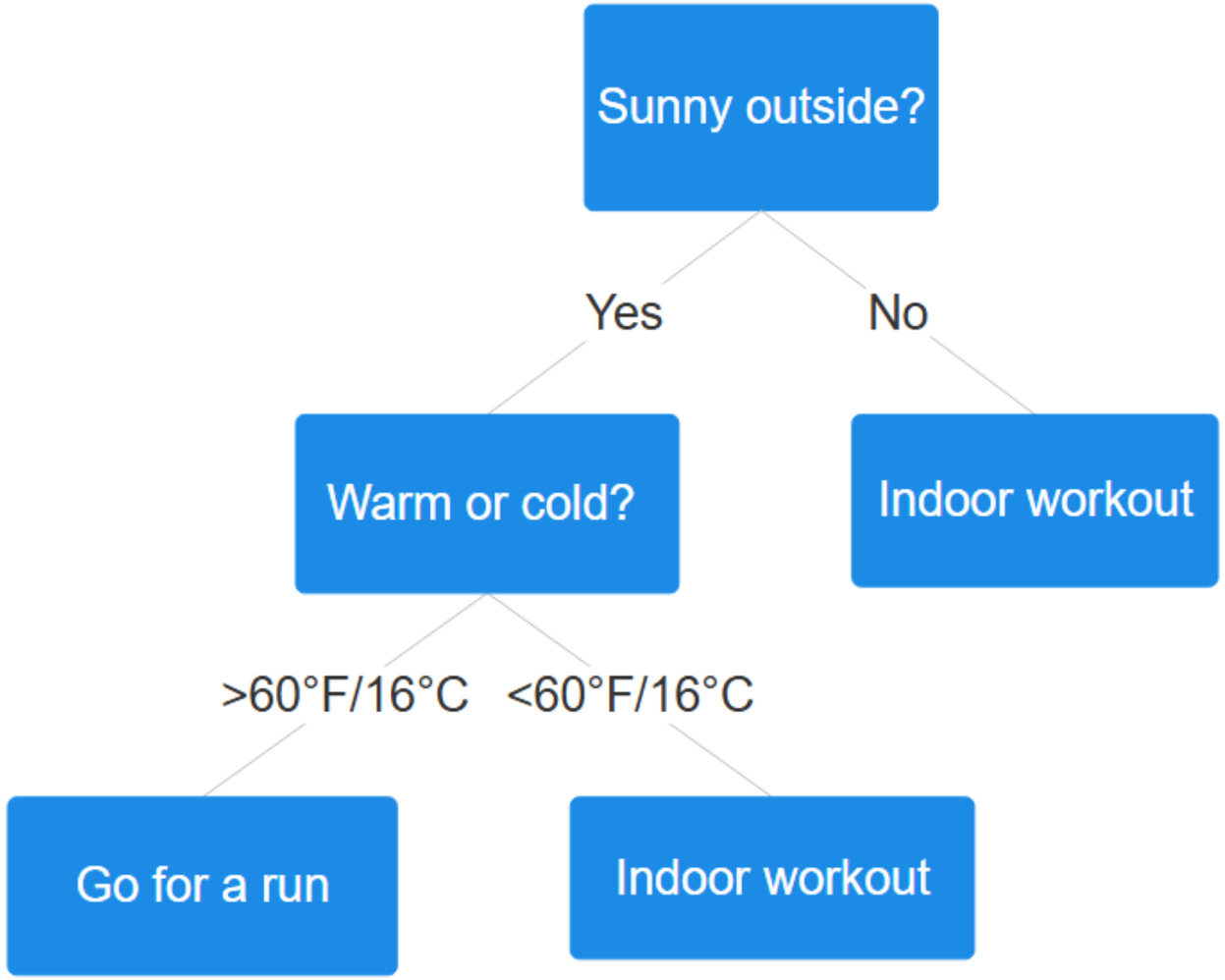
ADVANCED PROBABILITY: UNCERTAINTY IN DATA



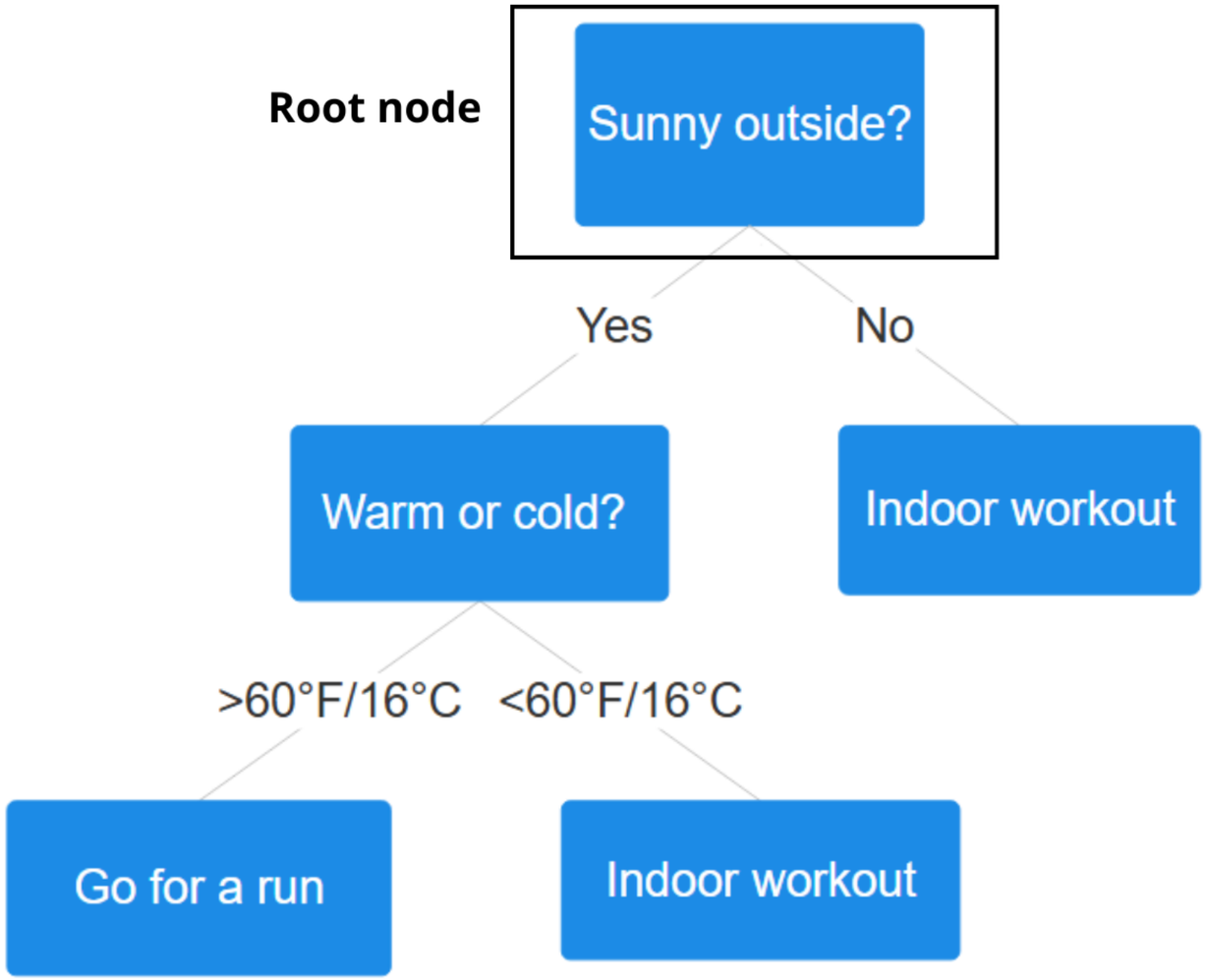
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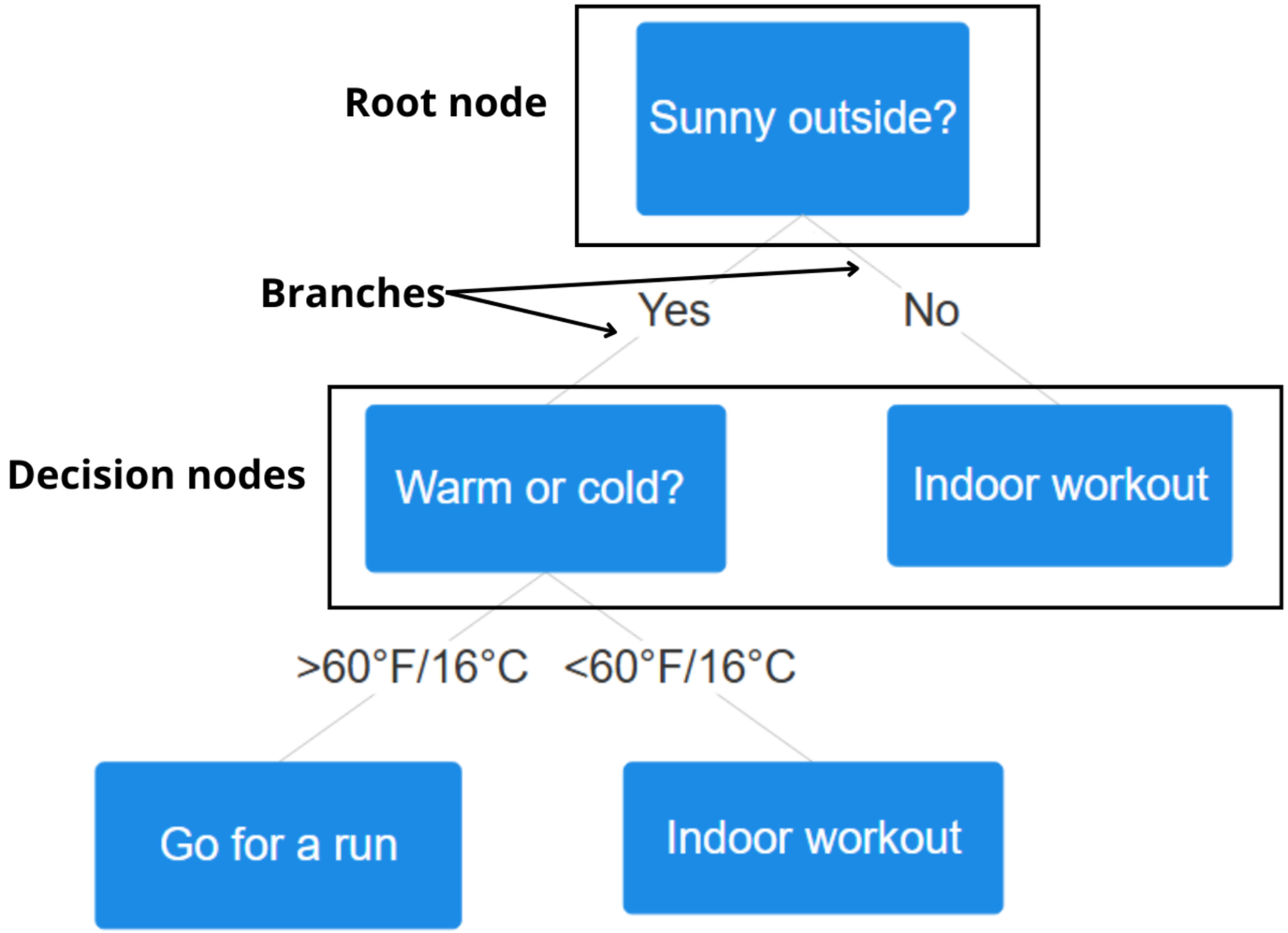
What is a decision tree?



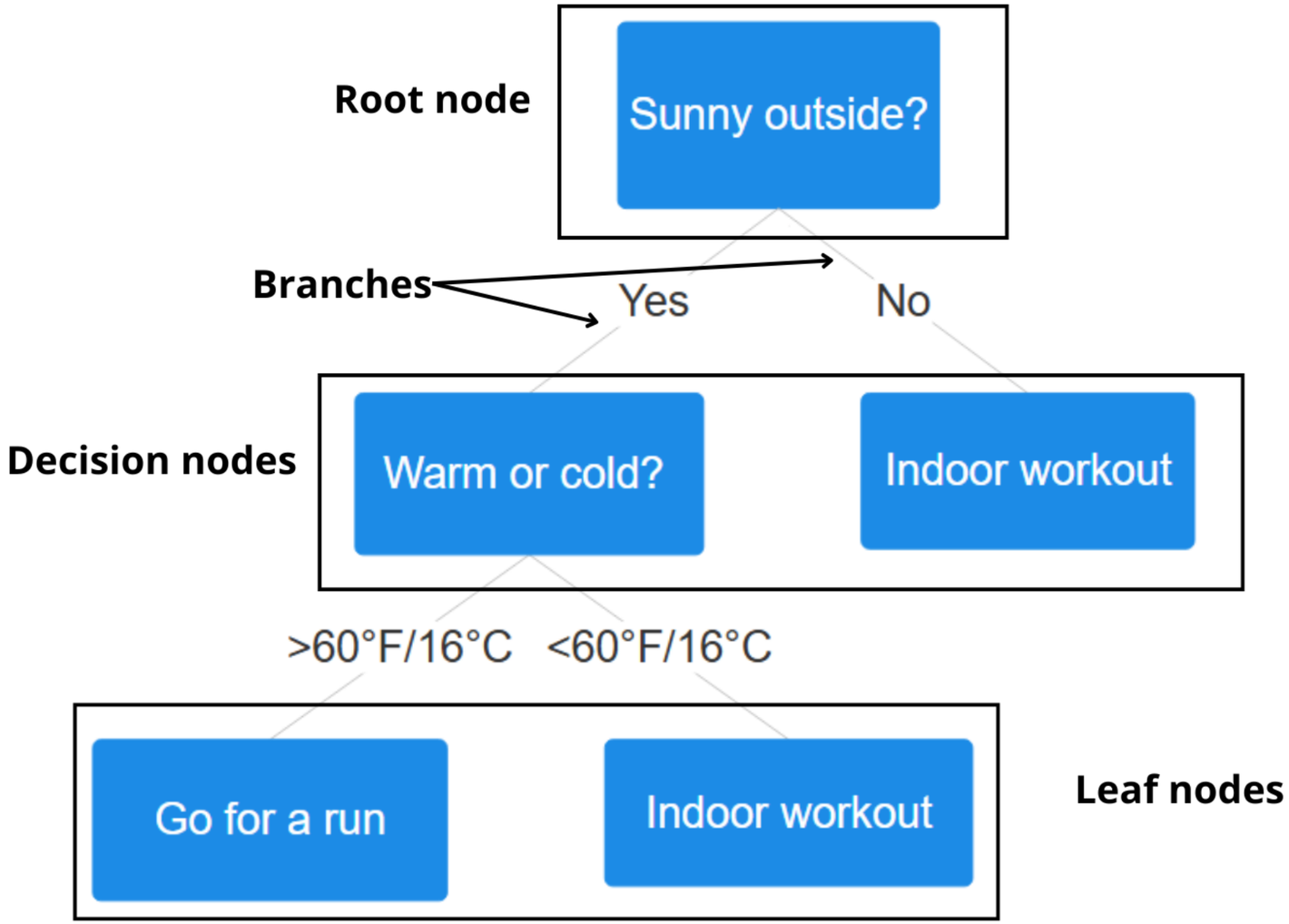
Building blocks of a decision tree



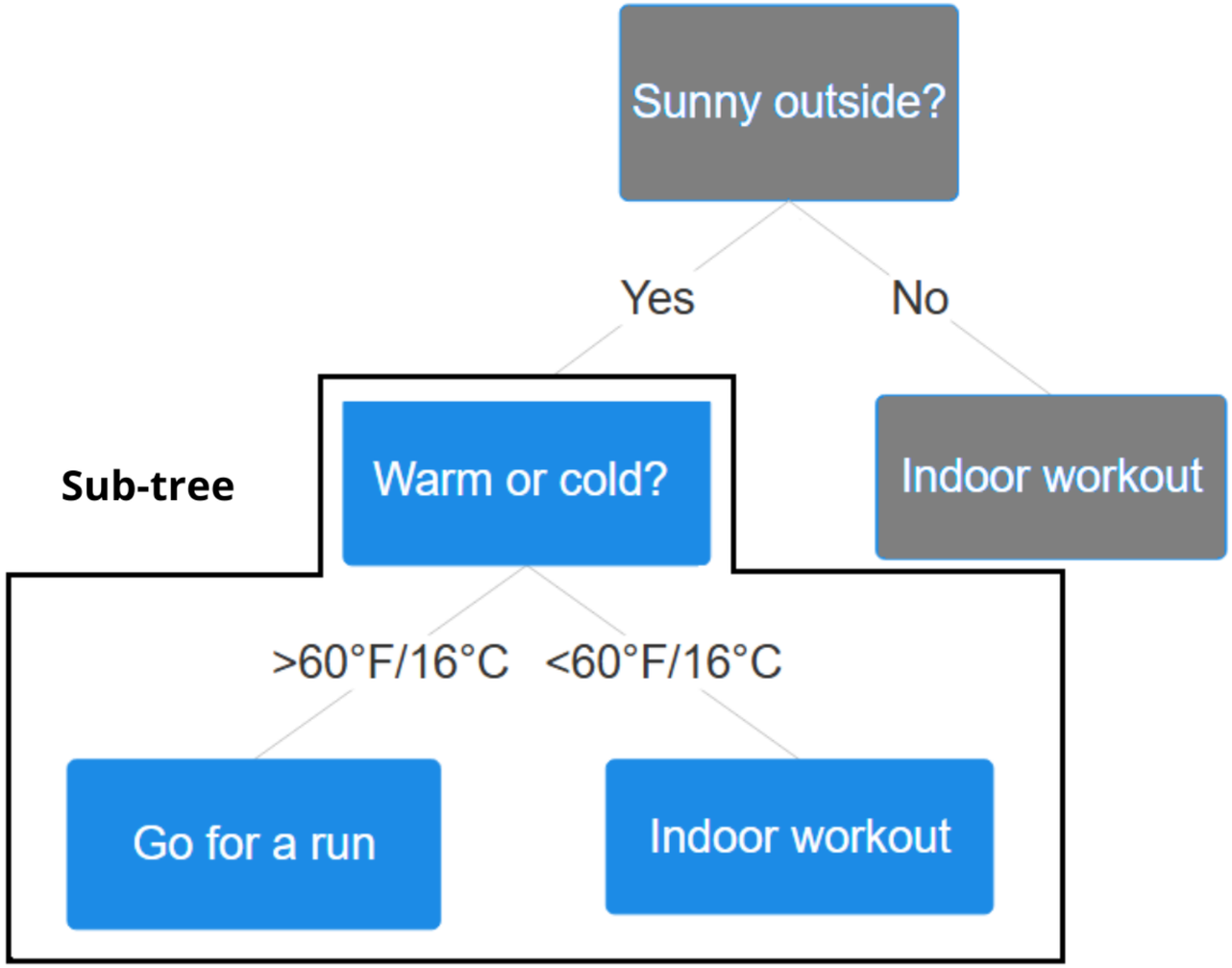
Building blocks of a decision tree



Building blocks of a decision tree



Building blocks of a decision tree

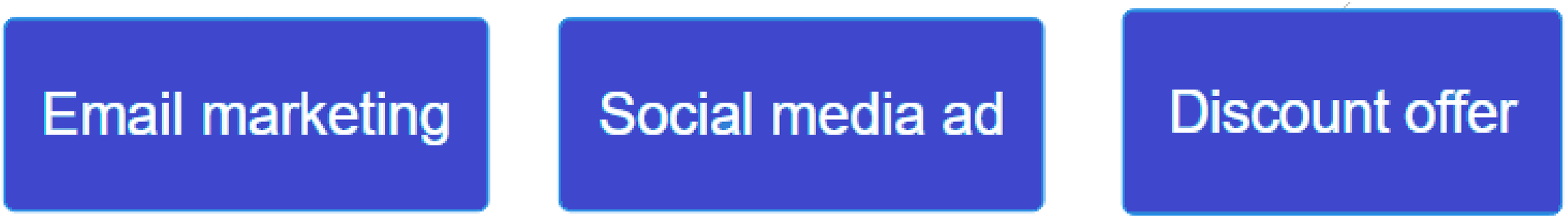


How decision trees work

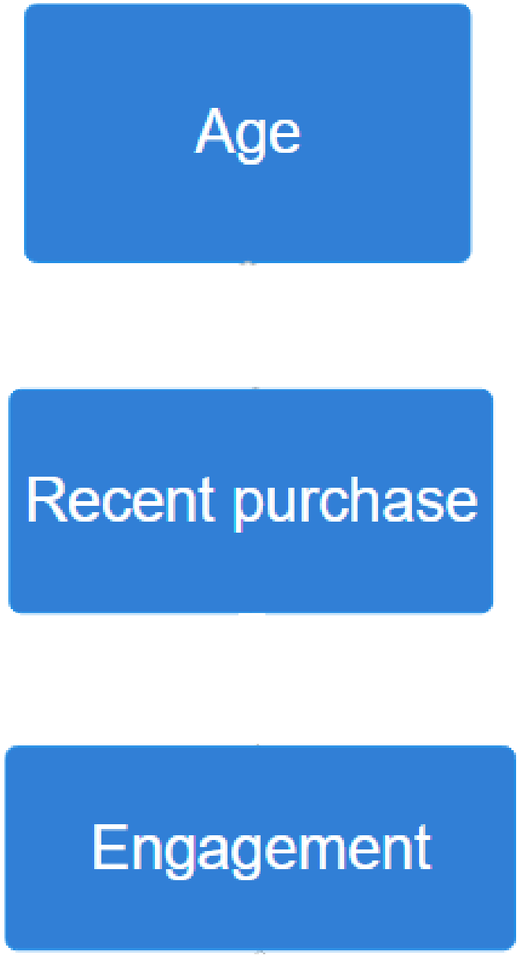
- Calculate based on data patterns
- Make splits using:
  - Statistical measures
  - `IF` - `THEN` logic
  - Probabilities
- Combine decision trees into random forests for more predictive power



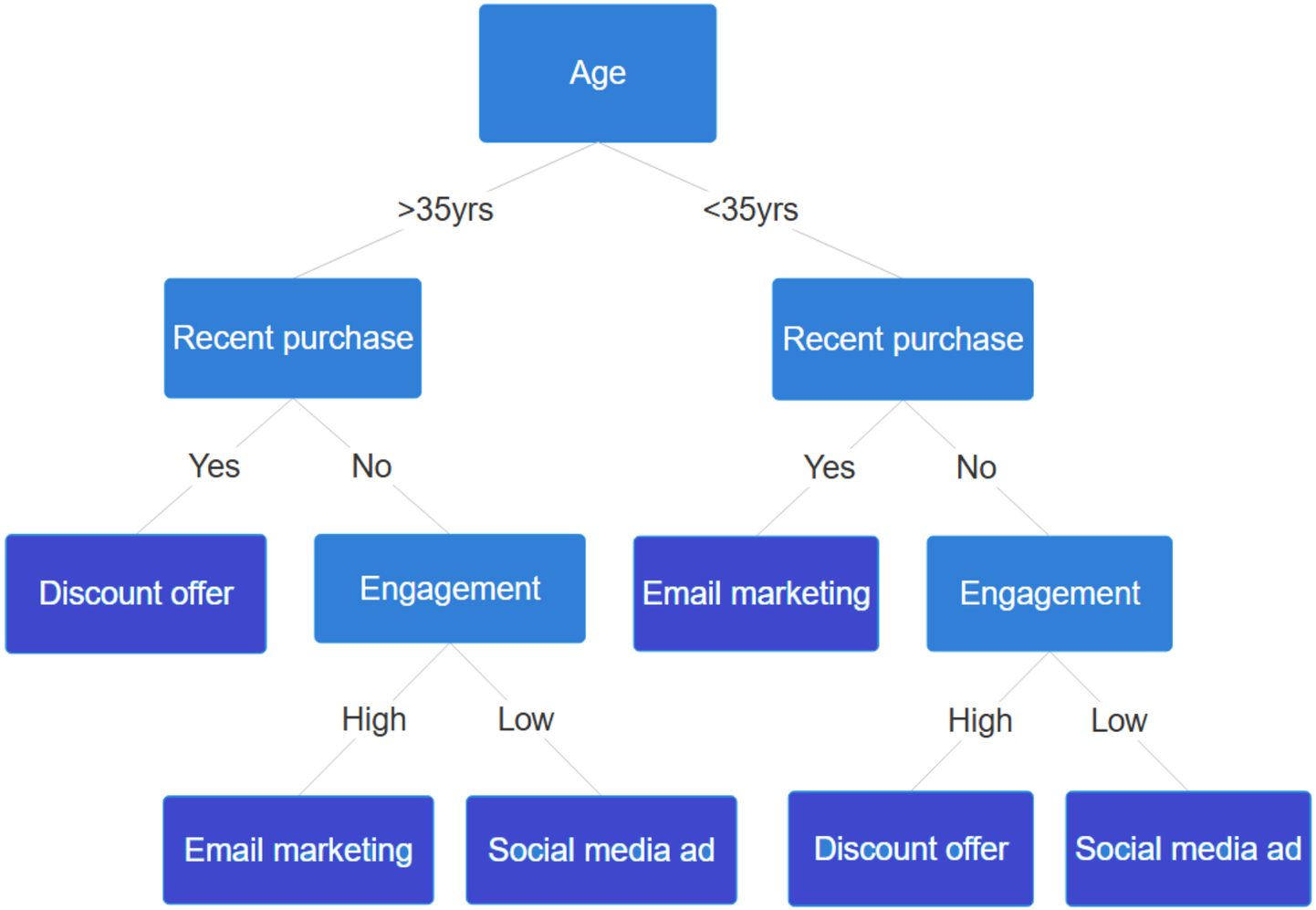
Example: choosing the right marketing strategy



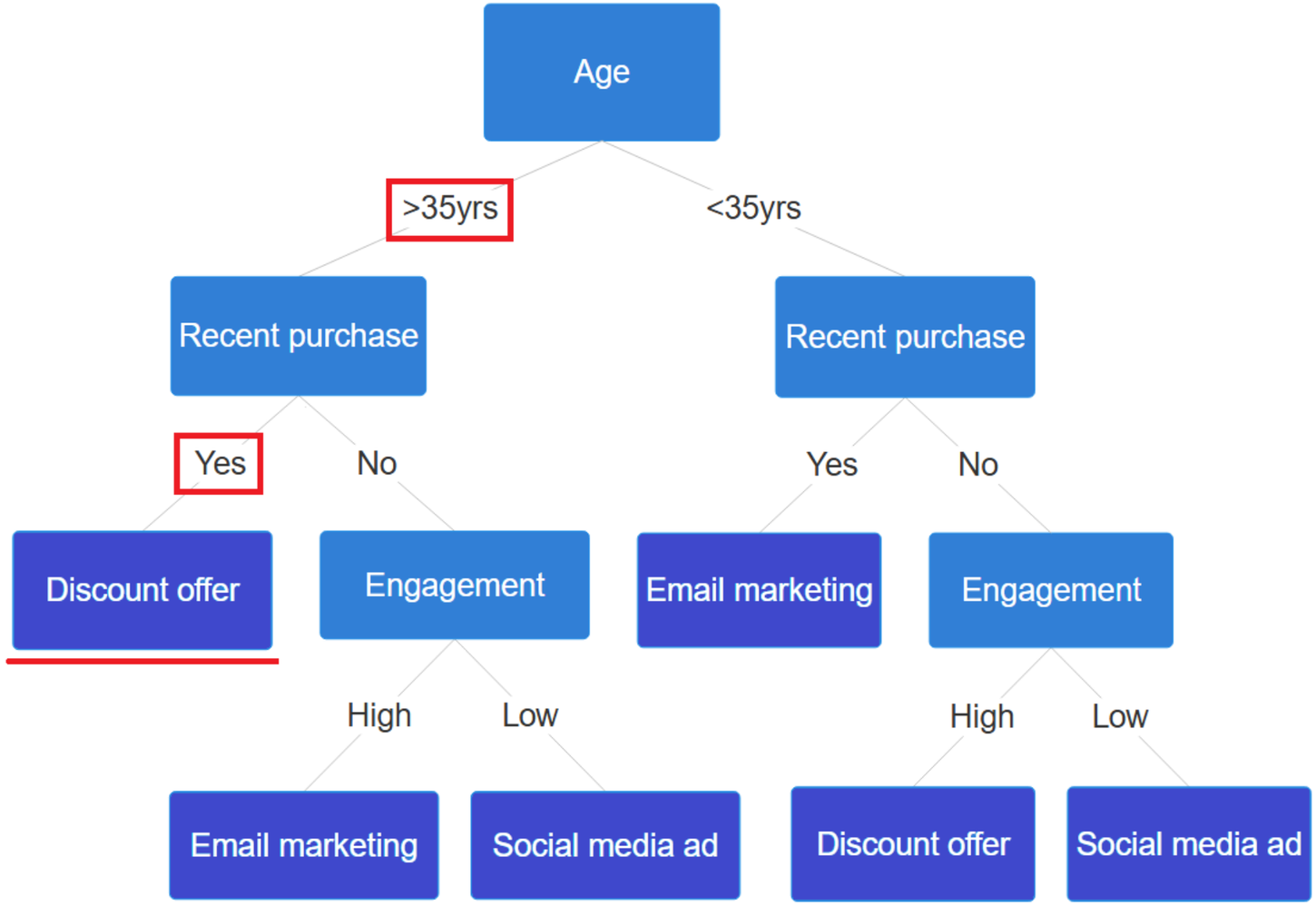
Example: choosing the right marketing strategy



Example: choosing the right marketing strategy



Example: choosing the right marketing strategy



Let's practice!

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Integrating insights in the decision process

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Decision-making under uncertainty

- Various ways to handle uncertainty
- How to integrate into decision-making?



Integrating insights for decision-making

1. Define the business question or problem



Integrating insights for decision-making

1. Define the business question or problem
2. Identify uncertainty factors



Integrating insights for decision-making

1. Define the business question or problem
2. Identify uncertainty factors
3. Choose the right method



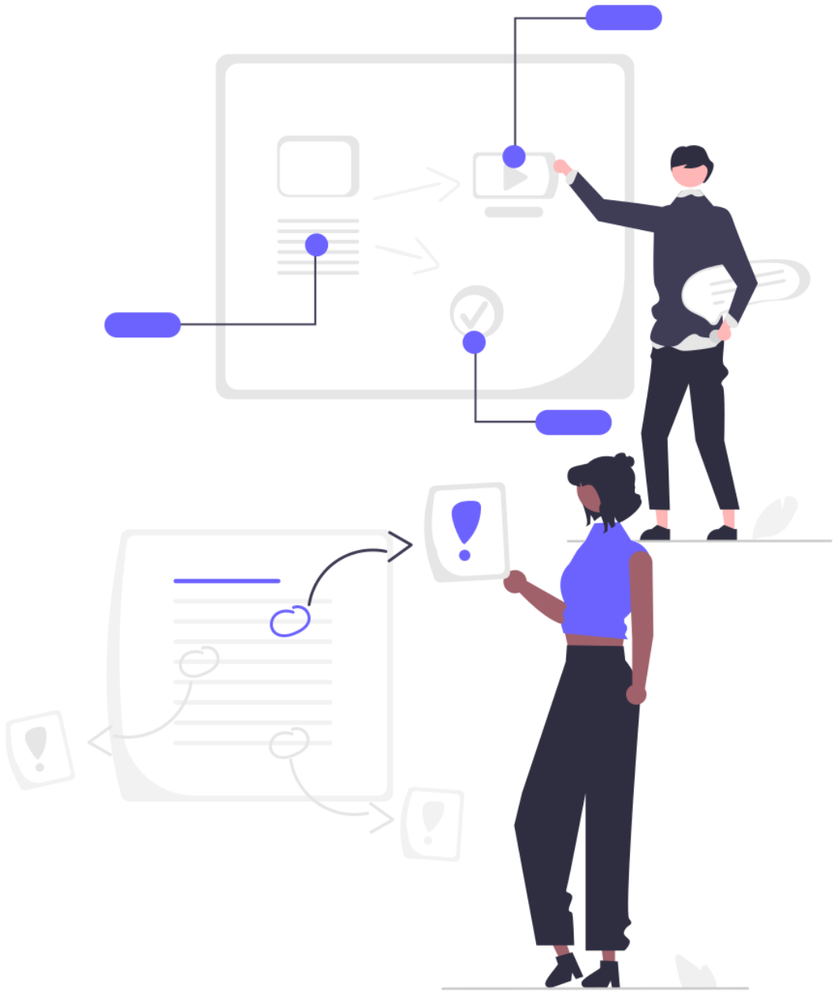
Integrating insights for decision-making

1. Define the business question or problem
2. Identify uncertainty factors
3. Choose the right method
4. Analyze and interpret the results



Integrating insights for decision-making

1. Define the business question or problem
2. Identify uncertainty factors
3. Choose the right method
4. Analyze and interpret the results
5. Communicate your insights



Example: expanding into a new market

- **Scenario:** expanding into a new market
- **Business question:** is expansion viable?
- **Uncertainties:** customer demand, costs, competitors



Example: expanding into a new market

- Choose a method: Monte Carlo simulation
  - Simulate variable outcomes
  - Assess likelihood of different levels of profitability
- Supplement with other methods to view the problem from another angle



Example: expanding into a new market

- Analyze results
  - 60% of simulations profitable
  - 20% break-even
  - 20% loss
- Visualize probability distribution and summarize metrics
- Decision: start with a pilot program



Let's practice!

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Wrap-up

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Recap

Chapter 1

- Multivariate distributions
- Conditional probability and Bayes' theorem
- Markov Chain analysis

Chapter 2

- Expected value calculation
- Confidence and prediction intervals
- Scenario analysis
- Sensitivity analysis

Chapter 3

- Resampling methods
- Monte Carlo simulation
- Decision trees
- Integrating insights into decisions

Next steps

Suggested follow-up courses:

- Data-Driven Decision Making for Business
- Data Storytelling Concepts
- Data Science for Business
- Implementing AI solutions in Business

Other Probability and Statistics courses

Congratulations!

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